SMART STUDY PLANNER WITH AI SUGGESTIONS

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THE REPORT

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Dehiwala

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SMART STUDY PLANNER WITH AI SUGGESTIONS

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I would like to thank everyone that has supported and assisted in the completion of the project.

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Abstract

Smart Study Planner is an online program created in order to help university students plan their studies effectively and enhance their studying skills. Many students have difficulties in managing a number of subjects, homework assignments, examinations and personal study schedule through conventional approaches such as notebooks, paper planners or simple reminder programs. This usually results in poor time management, late submissions and low academic performance. Hence, the idea of this project is to design a centralized and smart system that will be helpful to students in organizing and tracking their academic activities.

Users can register in a safe way, manage subjects, add and edit tasks, build up their own study timetables, track their academic progress, receive intelligent study tips and view their schedules via an calendar. Moreover, there are such features as notifications and uploading PDF notes.

Smart Study Planner is a product of contemporary full-stack web development technologies. Specifically, React.js, HTML5, CSS3, JavaScript, and Material UI were used to create the front-end of the program, while Node.js, Express.js, and MongoDB were used as a database management system for creating its back-end. RESTful API was utilized to facilitate communication between the front-end and back-end components.

Thus, the application created proves effective in providing users with a convenient and structured environment for planning and managing their educational activities. This tool facilitates time management, increases productivity, helps avoid missing important deadlines and allows tracking one's academic performance better. Smart Study Planner illustrates the way modern web development technologies can be applied to create a practical solution for students' education-related needs.

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CHAPTER 1

Introduction

1.1 Background of the Project

In today's fast-paced educational environment, university students are expected to manage multiple academic responsibilities, including lectures, assignments, examinations, presentations, projects and self-study activities. As the workload increases throughout the semester, many students find it difficult to organize their schedules effectively. Poor time management often results in missed deadlines, incomplete assignments, increased stress levels and reduced academic performance.

Although many students continue to use notebooks, paper diaries or simple reminder applications to manage their academic work, these traditional methods provide limited functionality and do not offer intelligent assistance for effective study planning. Existing digital tools often focus on only one aspect of academic management, such as note-taking or task reminders, rather than providing a complete academic planning solution.

To address these challenges, the Smart Study Planner was developed as a web-based academic management system that enables students to organize their academic activities through a single platform. The system provides facilities for managing subjects, scheduling tasks, generating personalized study timetables, monitoring study progress, viewing academic activities through a calendar interface, receiving intelligent study suggestions, managing PDF study materials and receiving notifications for important academic events.

The system was developed using modern full-stack web technologies including React.js, Node.js, Express.js, and MongoDB to provide a responsive, secure and user-friendly application. By integrating multiple academic management features into one platform, the Smart Study Planner aims to improve students' productivity, time management and overall academic performance.

1.2 Problem Statement

Most of the university students have troubles in managing their academic activities due to the usage of various tools like notebooks, calendar, reminder applications and messaging services. Since the mentioned tools are not connected, it becomes quite possible to forget about such things as assignment deadlines, study schedule and other academic activities

Besides that, the majority of currently existing applications for studying planning do not include all necessary features in one program. Most of them lack features like automatic timetable creation, studying process monitoring, personal recommendations for studying and adequate managing of subjects. Due to this, university students have to spend additional time on managing their activities rather than on studying itself.

Also, there is no single platform which can provide students with the opportunity to manage subjects, assignments, studying materials, notifications and general academic activities. As a result, using several separate applications leads to low efficiency of academic planning and even to the loss of important tasks.

In order to solve mentioned problems, this project suggests creating of Smart Study Planner, a web-based application which will provide users with the possibility of organizing their academic activities in one convenient place.

1.3 Objectives

1.3.1 General Objectives

To develop a web-based Smart Study Planner that helps university students effectively manage their academic activities, improve time management and increase study productivity.

1.3.2 Specific Objectives

- To develop a secure user registration and authentication system.
- To provide an interface for managing academic subjects.
- To enable students to create, update and organize study tasks.
- To generate personalized study timetables automatically based on user preferences.
- To provide a calendar view for managing academic schedules.
- To monitor study progress through visual progress tracking.
- To generate intelligent study suggestions based on task completion and subject priorities.
- To allow students to upload and manage PDF study materials.
- To notify users about upcoming deadlines and important academic activities.
- To develop a responsive and user-friendly web application using modern web technologies.

1.4 Scope of the Project

The scope of the Smart Study Planner is restricted to facilitating university students in managing their academic activities using a web application. The system allows registration and login of the users, management of subjects, generation of study tasks, uploading of study material in PDF format, generating personalized study timetables, tracking academic progress, getting intelligent study recommendations and scheduling via calendar.

The scope of the application is limited to personal study management rather than offering any institutional academic management solutions. The application does not offer any facilities for online examinations, learning management, student grading or lecturer management. The application only provides the facilities to help the students in organizing their studies.

1.5 Limitations

- The system is designed primarily for individual university students and does not support multiple user roles such as lecturers or administrators.
- AI suggestions are generated using predefined logical rules rather than advanced machine learning algorithms.
- The timetable generation process follows predefined scheduling logic and does not optimize schedules using artificial intelligence techniques.
- The system currently operates as a web application and does not include dedicated Android or iOS mobile applications.
- Notification functionality is limited to the system interface and does not currently support email or SMS notifications.
- Internet connectivity is required to access all system features because the application does not provide offline functionality.
- The system has been tested using a local MongoDB database and localhost deployment environment.

CHAPTER 2

Literature Review

2.1 Existing Systems or Similar Applications

A variety of planning tools are currently available for students to use. Some of the most common planning applications include Google Calendar, Microsoft To Do, Notion, Todoist, MyStudyLife, and Trello.

- One of the most popular tools that can be used for planning by students is Google Calendar. This tool allows you to create events, set reminders and plan your daily schedule. Nevertheless, Google Calendar is mainly a calendar tool that does not have features such as subject management, progress monitoring or recommendations about studying.
- Microsoft To Do is a task management application that helps you create your own task lists and reminders. Even though it gives an opportunity to easily manage your activities, it cannot generate your timetable automatically or monitor your subjects' performance.
- Notion offers a comprehensive work environment that integrates notes, tasks and database. Despite being highly flexible, users have to develop a study plan on their own, which can be quite difficult and time-consuming.
- MyStudyLife is a platform dedicated to students that helps organize class schedule, tasks and exams. Nonetheless, it does not offer any intelligent study recommendations, visual progress tracking and PDF resources management.
- Trello is a project management software that utilizes the Kanban approach to organizing work. It is commonly used by students to organize assignments and projects, yet it is not focused on studying and does not generate schedules automatically.

2.2 Technologies used in the Industry

Modern web-based academic management systems are commonly developed using full-stack web technologies that provide scalability, responsiveness, and efficient data management.

- React.js, Angular and Vue.js are common frameworks for frontend development that can be used to build an engaging and highly responsive interface.
- Node.js, Express.js, Spring Boot and Django are popular frameworks for backend development and are also good choices for building RESTful API and efficient server-side operations.
- Commonly used database management systems for managing user data, grades, schedule and task information include MongoDB, MySQL, and PostgreSQL.
- JWT (JSON Web Token) and OAuth are common choices for user authentication.
- Common choices of modern UI frameworks are Material UI, Bootstrap and Tailwind CSS for developing visually appealing interfaces.
- Moreover, cloud storage and notification services are also used in academic applications for better user engagement.
- Smart Study Planner implements some of these modern technologies as it makes use of React.js, Node.js, Express.js, MongoDB, Material UI, HTML5, CSS3 and JavaScript to build an end-to-end full stack web application.

2.3 Comparison of Existing Solutions

Feature	Google Calendar	Microsoft To Do	MyStudyLife	Notion	Smart Study Planner
Subject Management	×	×	✓	✓	✓
Task Management	limited	✓	✓	✓	✓
Timetable Generation	×	×	✓	×	✓
Progress Tracking	×	×	×	limited	✓
AI Study Suggestions	×	×	×	limited	✓
Calendar View	✓	×	✓	✓	✓
PDF Notes Upload	×	×	×	✓	✓
Notifications	✓	✓	✓	✓	✓

Table 2.3 Comparison table

It is evident from the analysis that although there are a number of useful academic planning applications available, none of these integrate all the necessary functionality together. The Smart Study Planner integrates a number of functions like subject management, task management, automatic generation of time-tables, progress tracking, study suggestions based on AI technology, PDF management, calendar and notifications in one web application.

CHAPTER 3

System Analysis (SRS)

3.1 Functional Requirements

User Management

- ✓ User registration
- ✓ User login and authentication
- ✓ User profile management
- ✓ Secure logout

Subject Management

- ✓ Add subjects
- ✓ Edit subject details
- ✓ Delete subjects
- ✓ Store subject priority
- ✓ Store subject difficulty

Task Management

- ✓ Add study tasks
- ✓ Edit tasks
- ✓ Delete tasks
- ✓ Upload PDF notes
- ✓ Track task status
- ✓ Manage deadlines

Timetable Management

- ✓ Create study time slots
- ✓ Automatically generate study timetable
- ✓ Display timetable

Calendar Management

- ✓ Display monthly calendar
- ✓ Show upcoming deadlines

Progress Tracking

- ✓ Display completed tasks
- ✓ Display pending tasks
- ✓ Display Total tasks
- ✓ Display total Subjects
- ✓ Generate subject progress
- ✓ Display overall study progress

AI Suggestion Module

- ✓ Generate study suggestions
- ✓ Detect approaching deadlines
- ✓ Identify high-priority subjects
- ✓ Recommend hard subjects
- ✓ Analyse study progress

Notification Module

- ✓ Display notifications
- ✓ Alert users about deadlines
- ✓ Notify timetable events

3.2 Non-Functional Requirements

- Performance

  The system should load pages within a few seconds.

  Database operations should be performed efficiently.

- Reliability

  User information should be stored accurately.

  The system should recover gracefully from unexpected errors.

- Security

  User passwords should be encrypted.

  Unauthorized users should not access protected resources.

  User-specific data should remain private.

- Usability

  The interface should be simple and user-friendly.

  The application should be easy to learn and navigate.

- Maintainability

  Source code should follow modular programming principles.

  New features should be easily integrated.

- Scalability

  The system should support future expansion.

  Additional modules can be incorporated without major redesign.

3.3 Input/Output Requirements

3.3.1 Input Requirements

- User registration details
- Login credentials
- Subject information
- Subject priority
- Subject difficulty
- Task information
- Task deadlines
- Task status
- PDF study materials
- Timetable preferences
- Study time slots

3.3.2 Output Requirements

- User dashboard
- Subject list
- Task list
- Generated timetable
- Monthly calendar
- AI study suggestions
- Notifications
- Progress charts
- Subject progress bars
- Overall completion percentage

3.4 Hardware and Software Requirements

3.4.1 Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: 4 GB minimum (8 GB recommended)
- Storage: 20 GB available space
- Internet Connection
- Keyboard and Mouse

3.4.2 Software Requirements

Software	Purpose
Windows 10 / Windows 11	Operating System
Visual Studio Code	Code Editor
Node.js	Backend Runtime
React.js	Frontend Development
Express.js	Rest API Development
MongoDB	Database
MongoDB Compass	Database Management
Postman	API Testing
Git & GitHub	Version Control
Google Chrome	Web Browser

Table 3.2 Software requirements

CHAPTER 4

System Design

4.1 Use Case Diagram

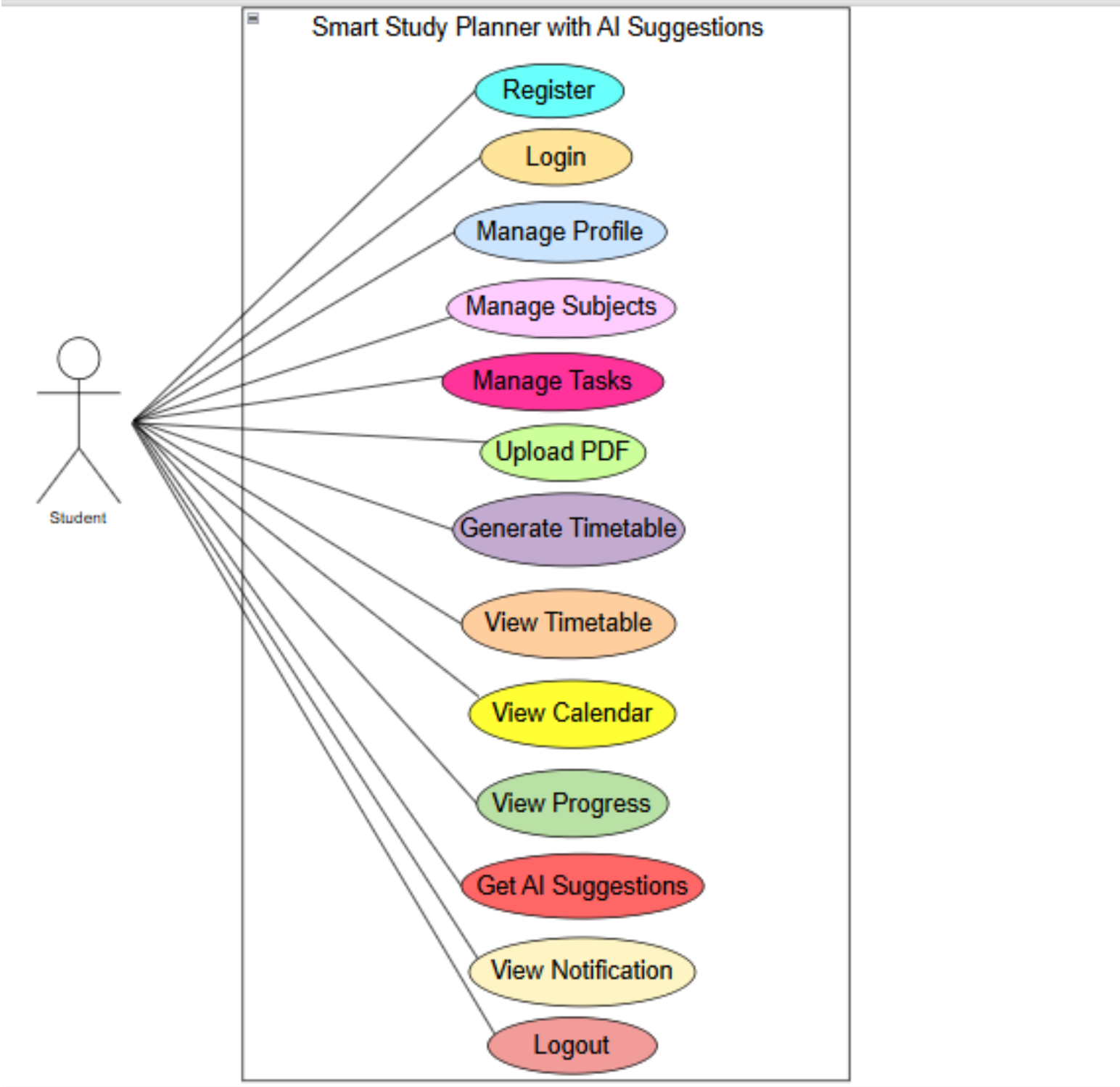


Figure 4.1 Use Case diagram

4.2 Activity Diagram

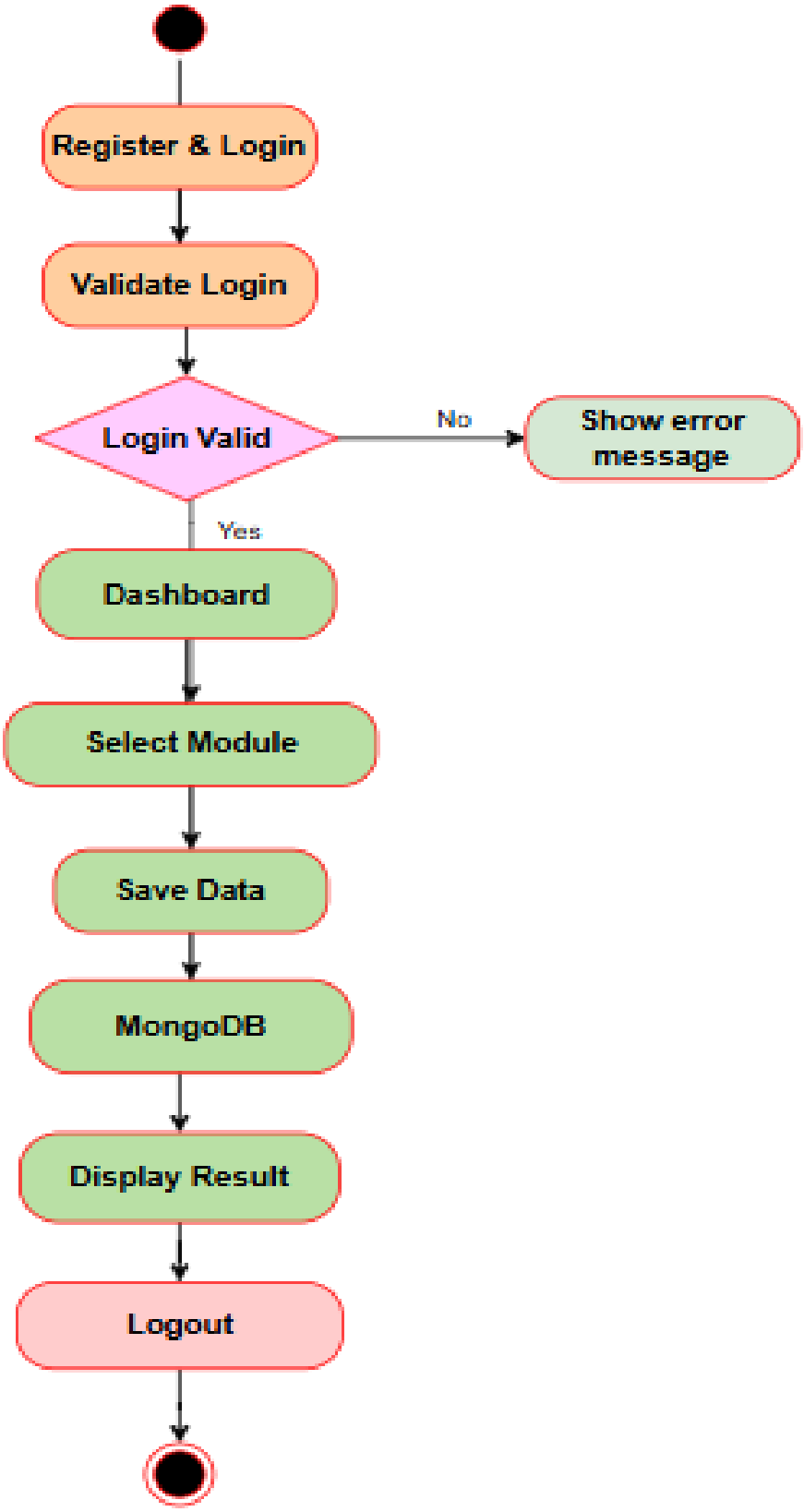
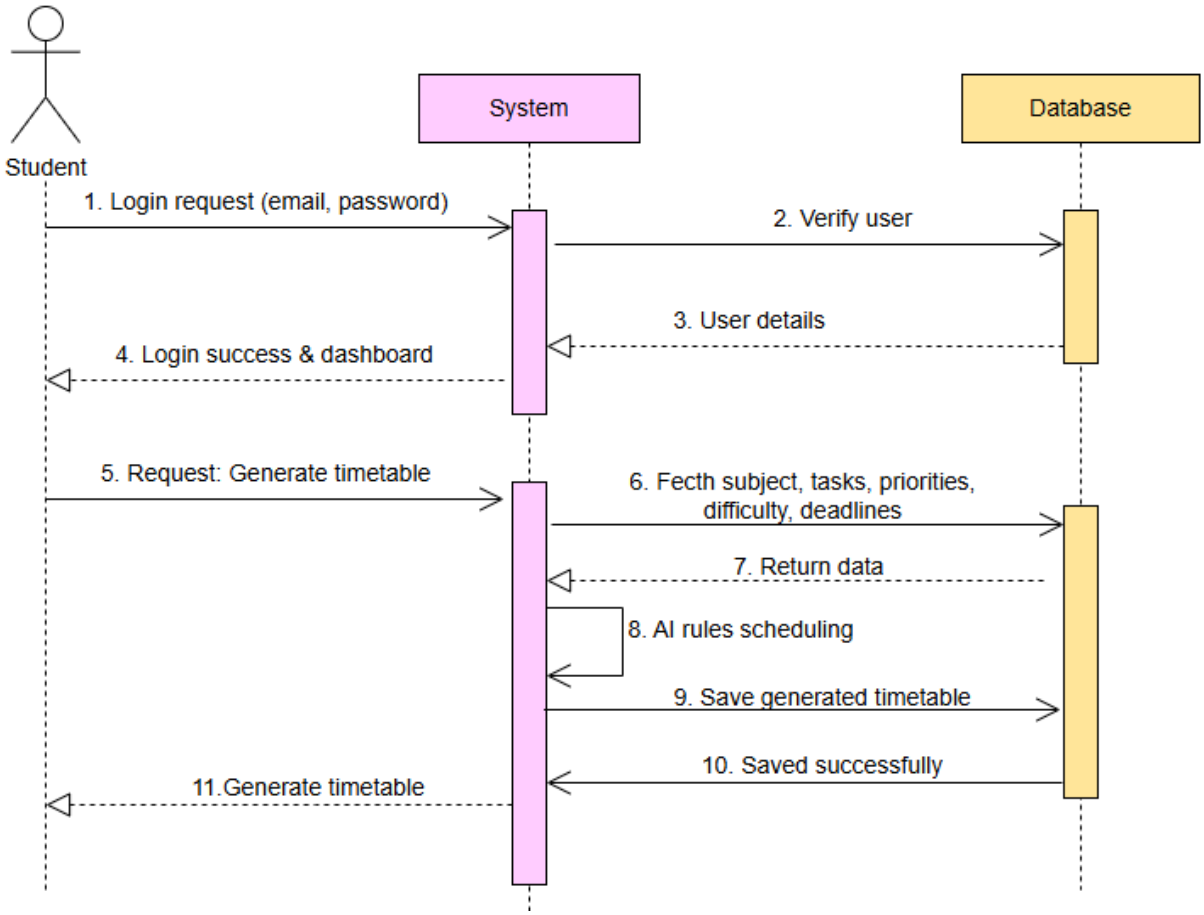


Figure 4.2 Activity diagram

4.3 Sequence Diagram



4.3 Sequence diagram

4.4 Class Diagram

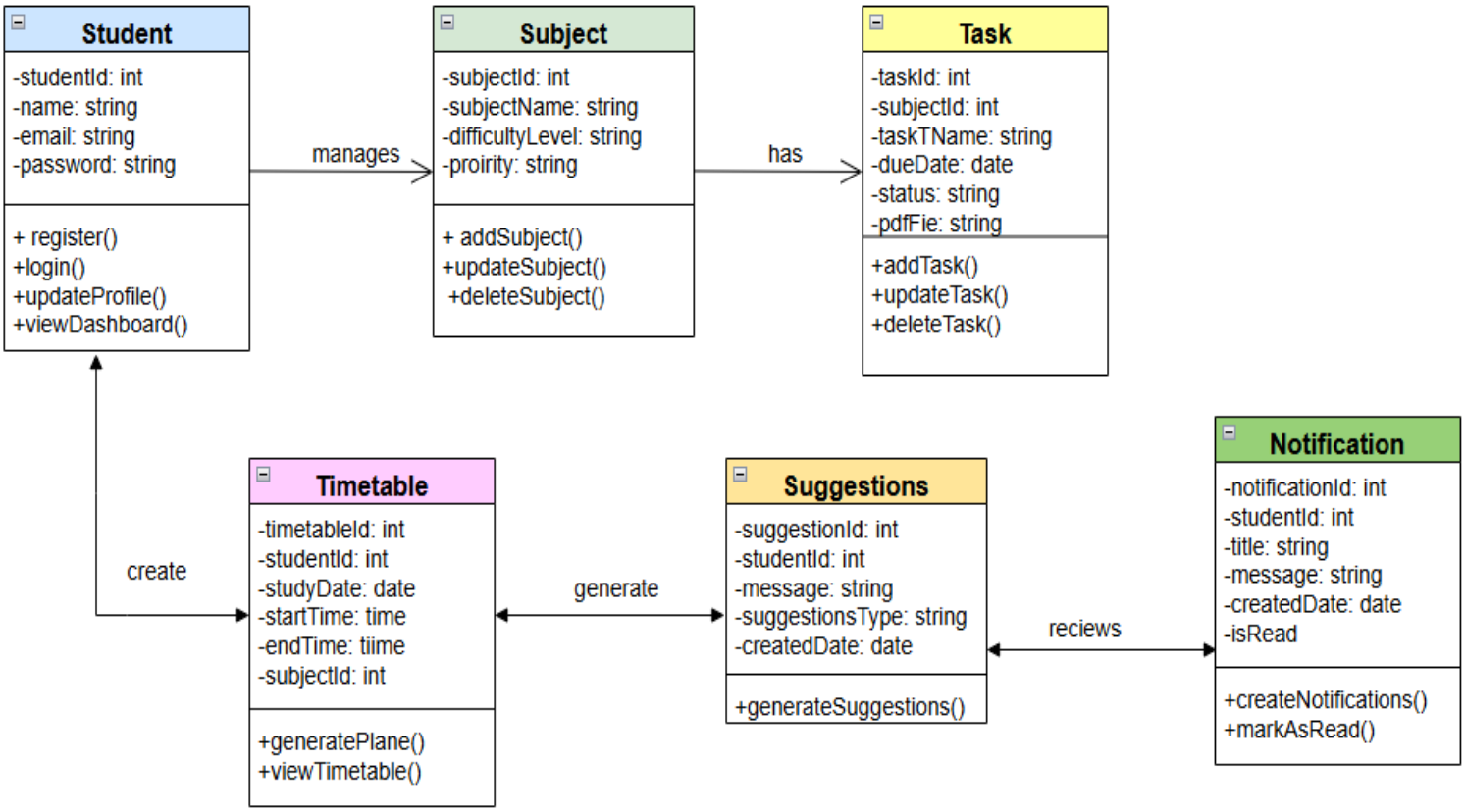


Figure 4.4 Class diagram

4.5 ER Diagram

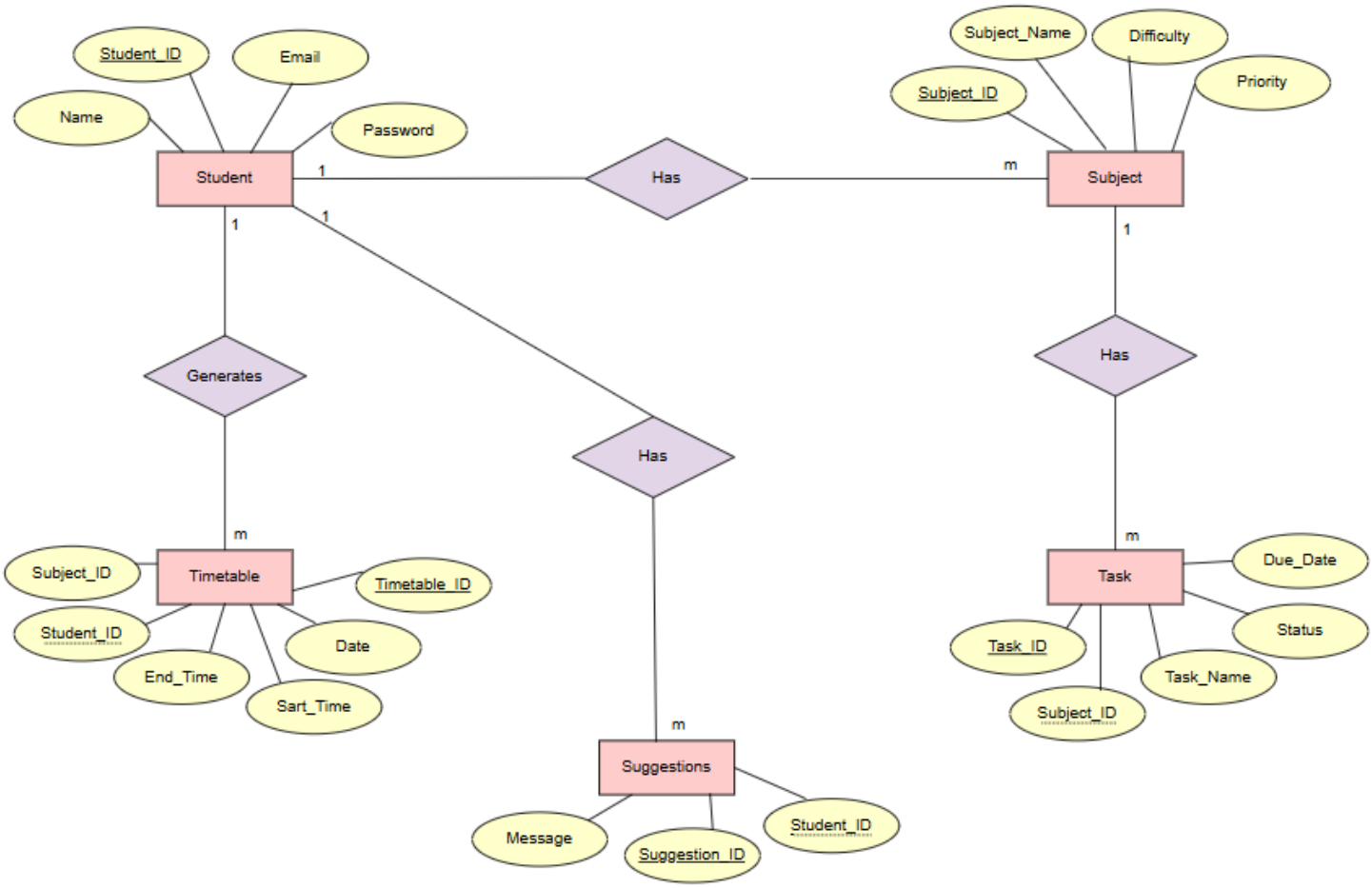


Figure 4.5 ER diagram

CHAPTER 5

Implementation

5.1 Programming Language Used

Programming Language	Purpose
JavaScript	Frontend and backend development
JSX	React component development
HTML5	Structure of web pages
CSS3	User Interface design and implement

Table 5.1 Programming language

JavaScript was selected as the primary programming language because it enables both client-side and server-side development, allowing seamless communication between different system components.

5.2 Tool and Technologies

Technology	Purpose
React.js	Frontend development
Node.js	Backend runtime environment
Express.js	RESTful API development
MongoDB	NoSQL database
Mongoose	MongoDB object modelling
Visual Studio Code	Source cord editor
Git & GitHub	Version control
Postman	API testing
Multer	PDF file upload handling
Material UI	User Interface icons
CSS3	Responsive user interface design

Table 5.2 Tools and technologies

The combination of these technologies enabled the development of a modern, responsive and scalable web application.

5.3 Key Features

User Authentication

- The system enables users to register and login securely into the application using email and password.

Subject Management

- The application allows students to create and modify study subjects and set priority levels and difficulty levels.

Task Management

- The application allows students to create study tasks, upload related PDF files, set deadline for tasks, modify task status and manage assignments.

Automatic Timetable Generation

- The system creates a personalized study timetable for the user through analysis of priority of subjects, subject difficulty, deadlines and study hours.

AI-Based Study Suggestions

- The application makes intelligent study suggestions for the user based on his performance and academic database. The AI suggestion module has the following features:
  → Performance enhancement suggestions
  → High priority subject alert
  → Hard subject suggestions
  → Deadline alert
  → Pending task alert

Progress Tracking

- Student can view total study completion percentage using circular progress indicator and subject wise progress bar.

Interactive Dashboard

- The dashboard gives an insight into today's study goals, tasks to do, advice from AI, study schedule and study statistics.

Calendar Module

- All the assignments, study and deadlines are shown on the calendar.

Notification System

- Notifications and reminders are set up automatically and shown in the notifications module.

5.4 Code Structure

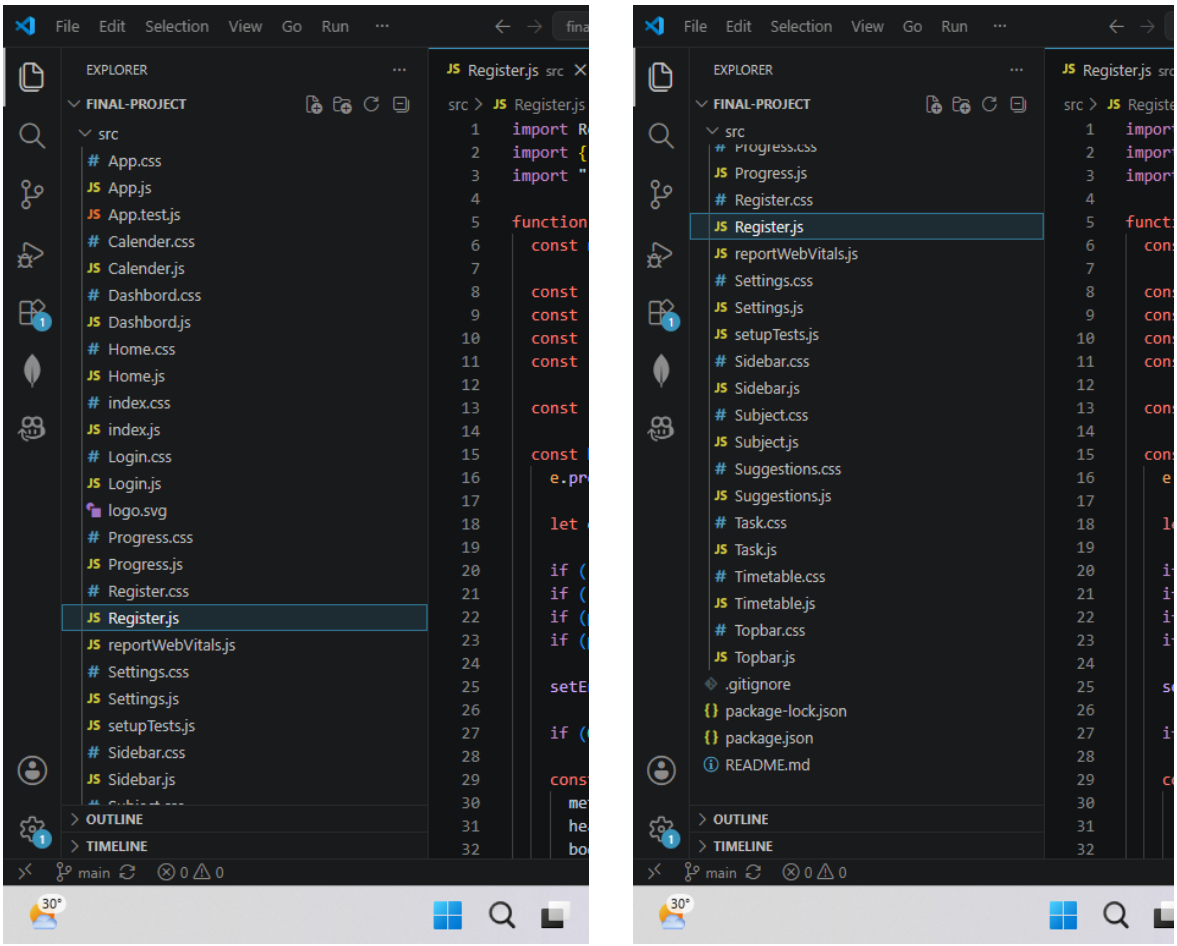


Figure 5.1 Code structure 1

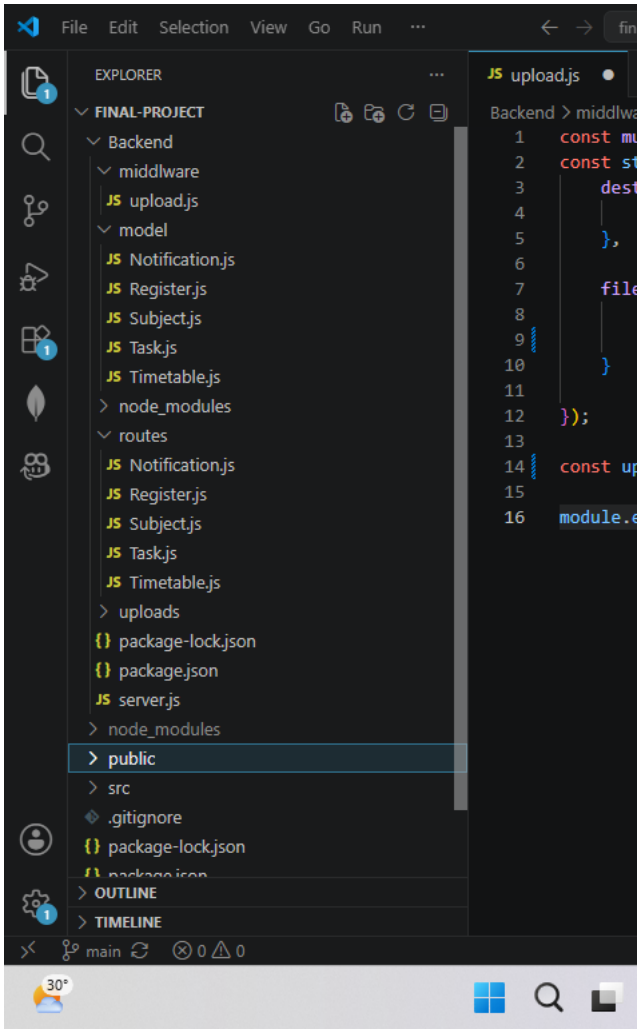


Figure 5.2 Code structure 2

5.5 GUI Screenshots

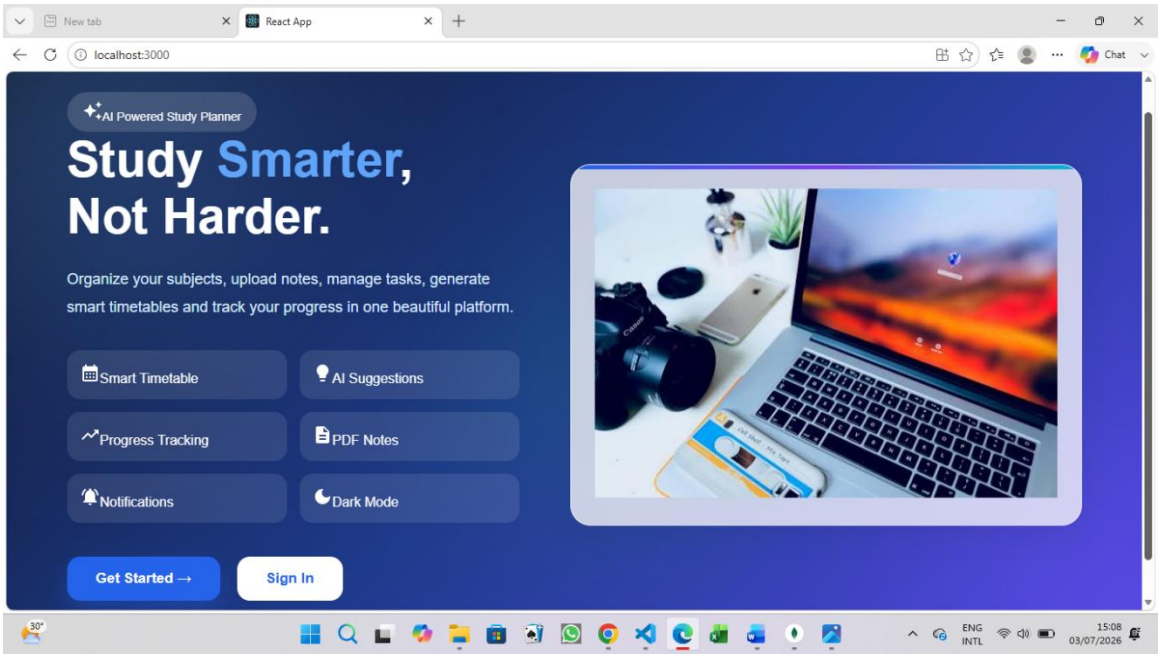


Figure 5.1 Home page

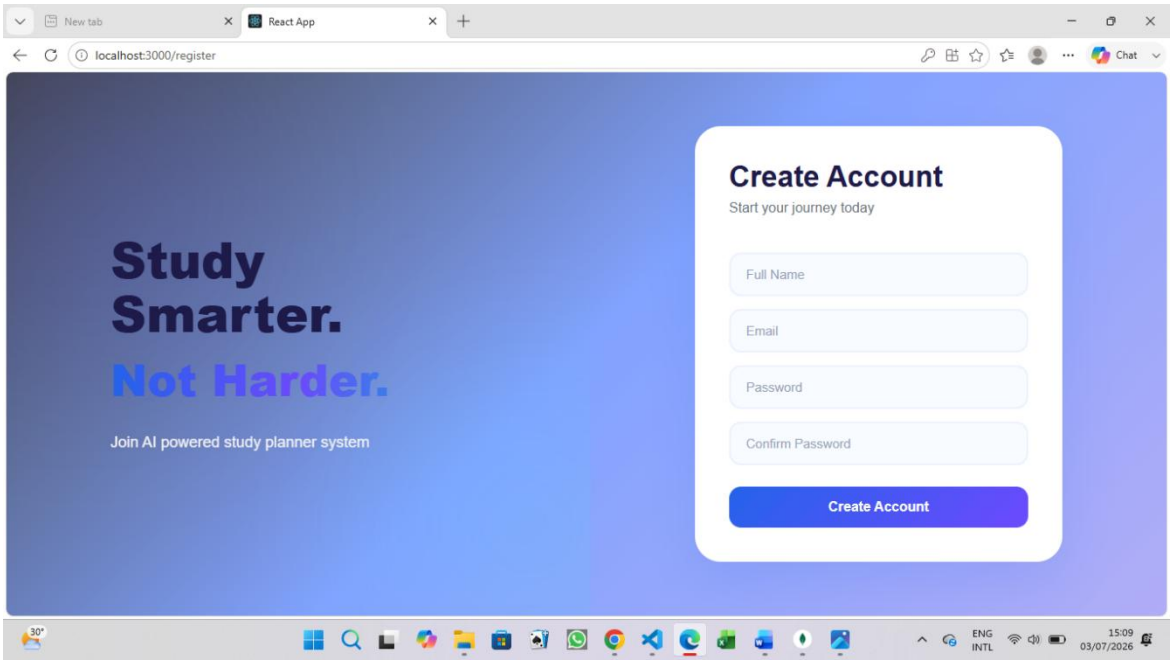


Figure 5.2 Register page

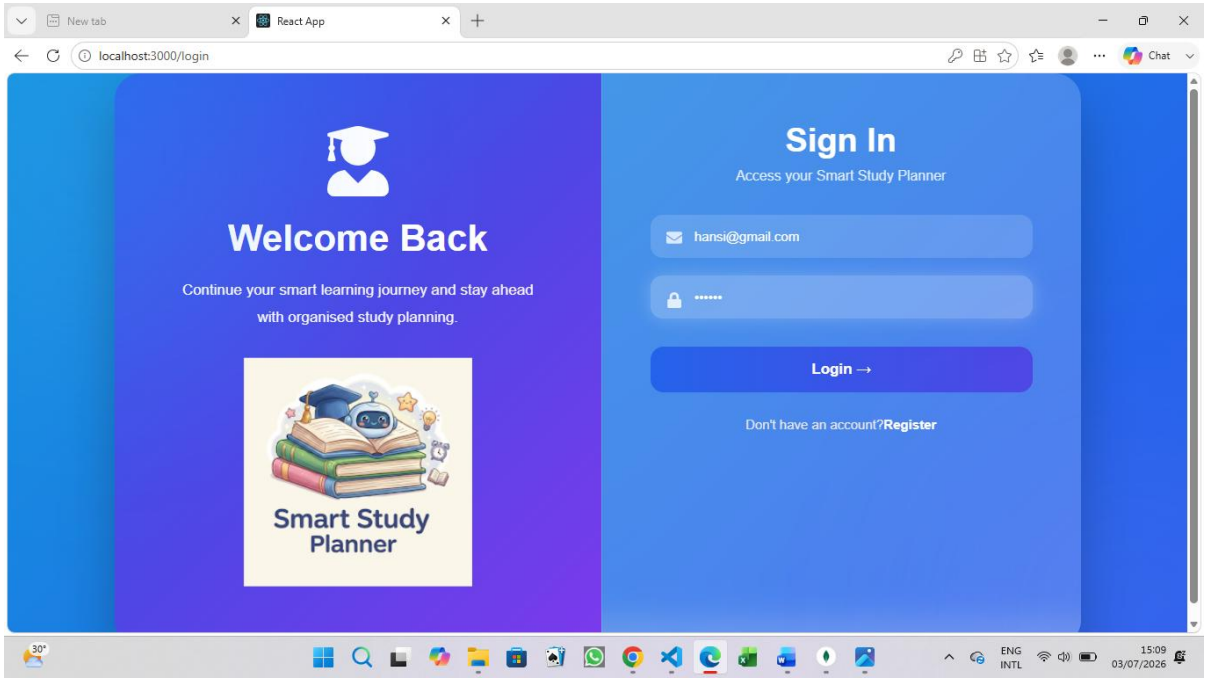


Figure 5.3 Login page

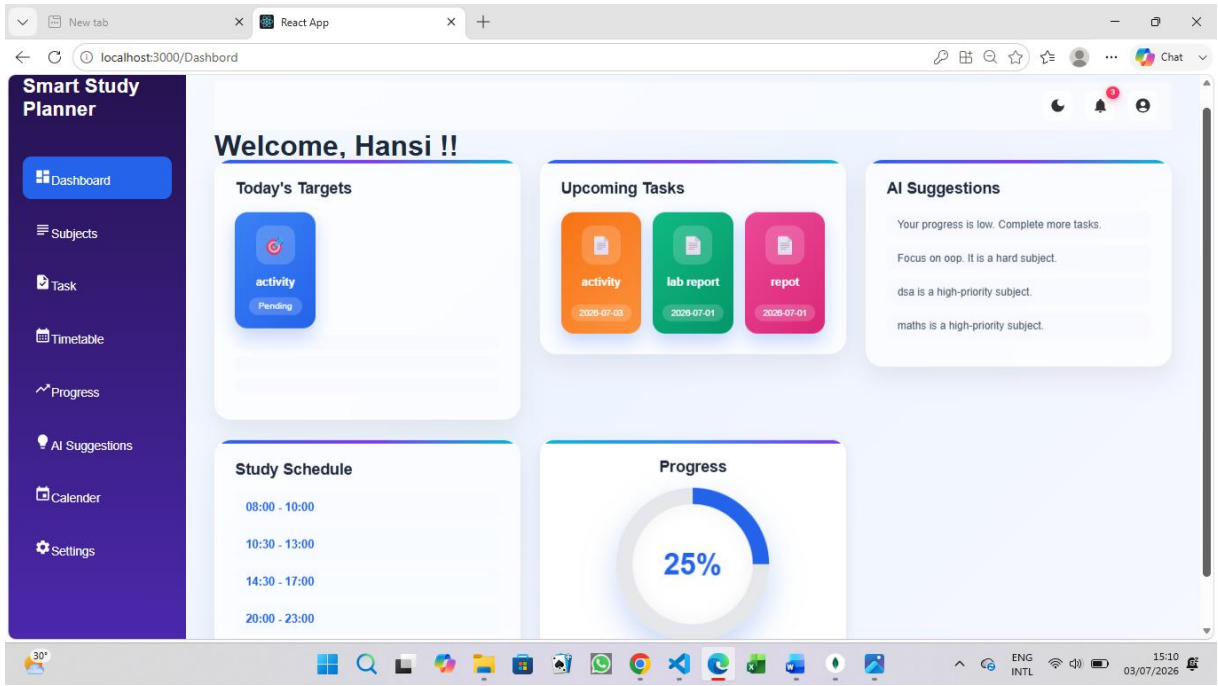


Figure 5.4 Dashboard page

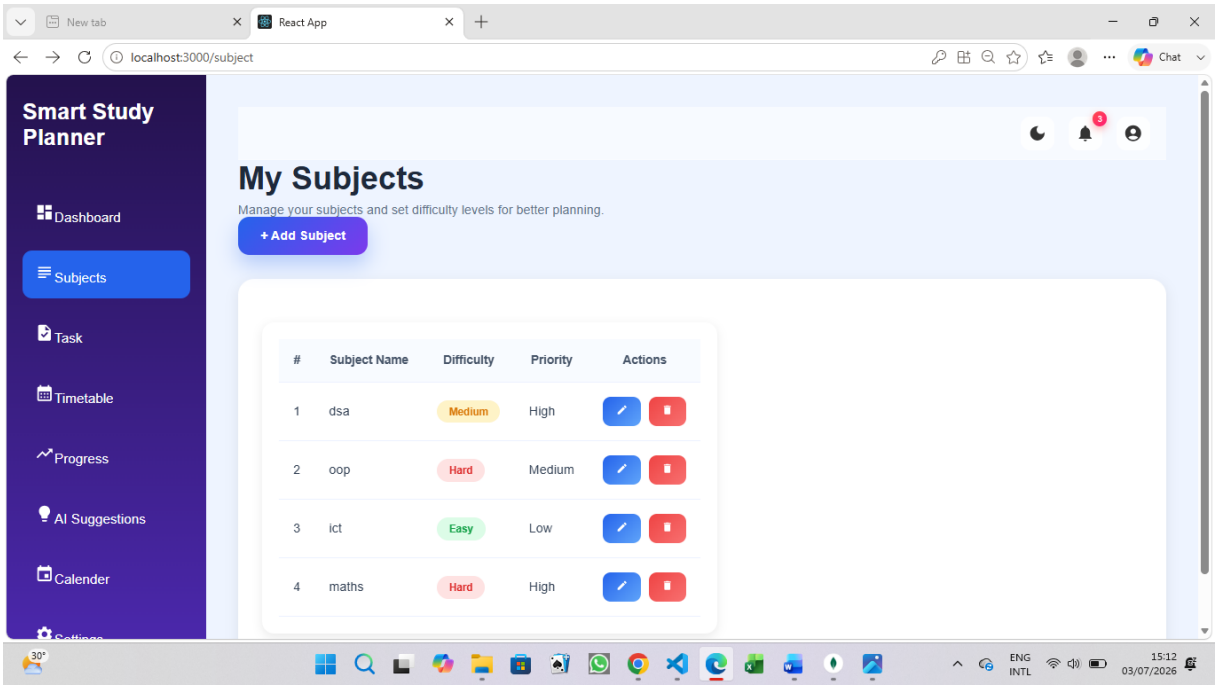


Figure 5.5 Subject page

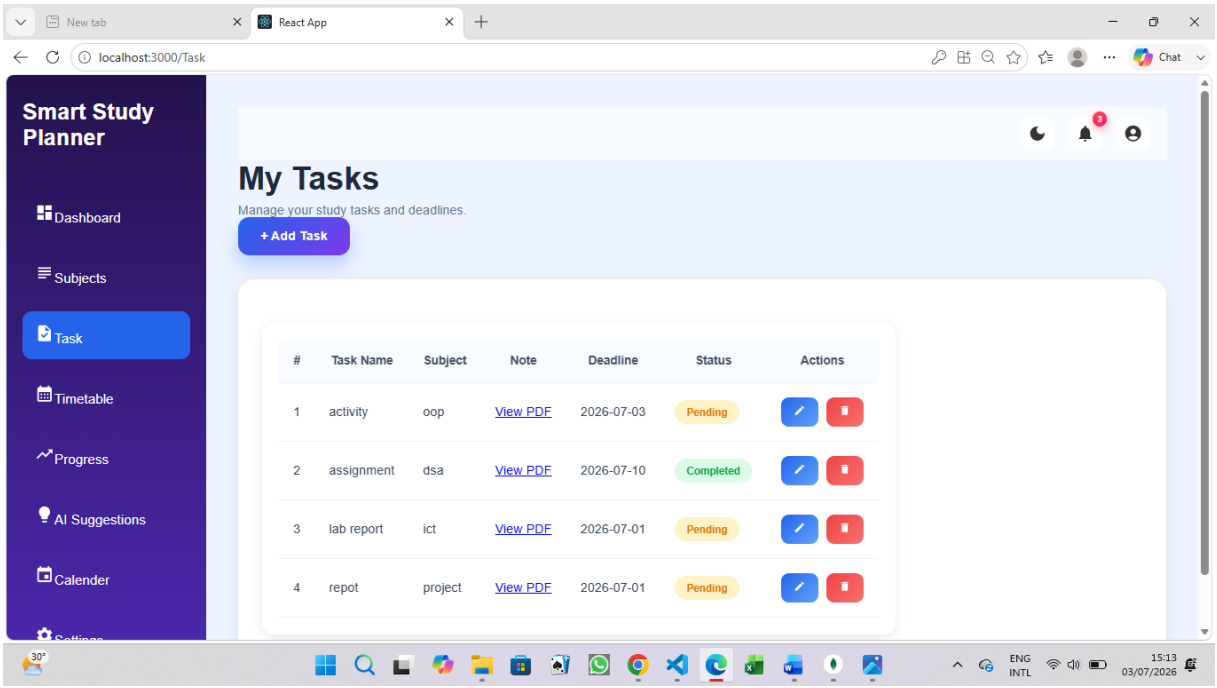


Figure 5.6 Task page

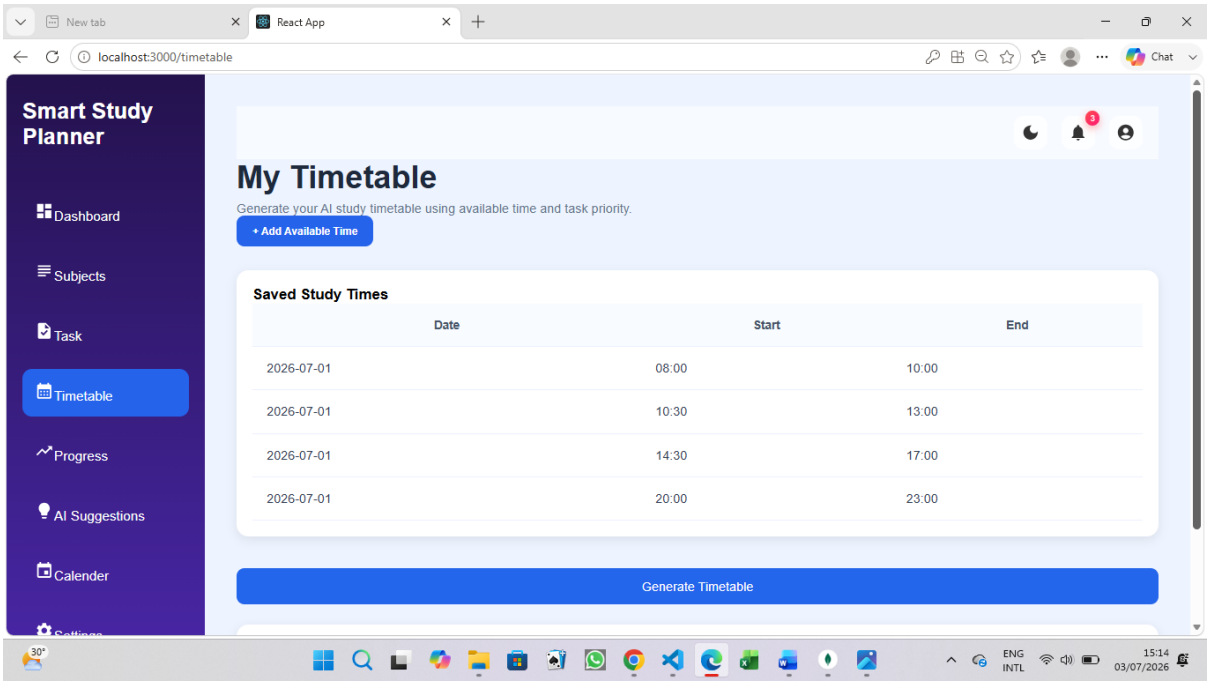


Figure 5.7 Timetable page

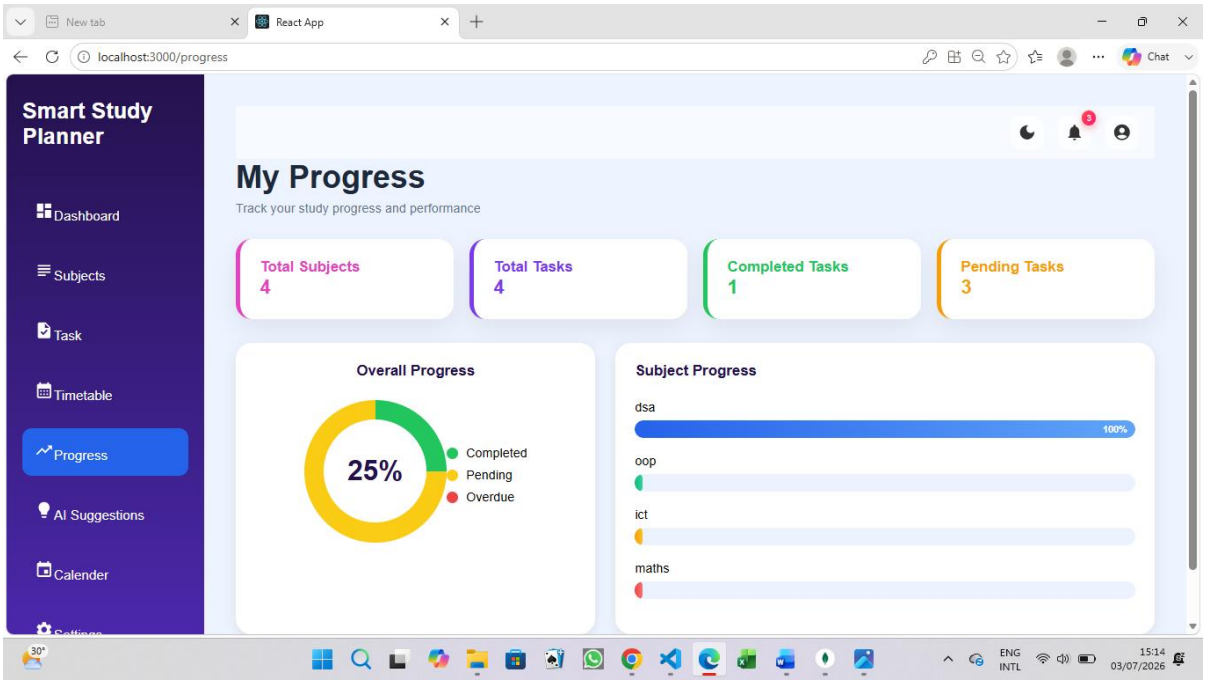


Figure 5.8 Progress page

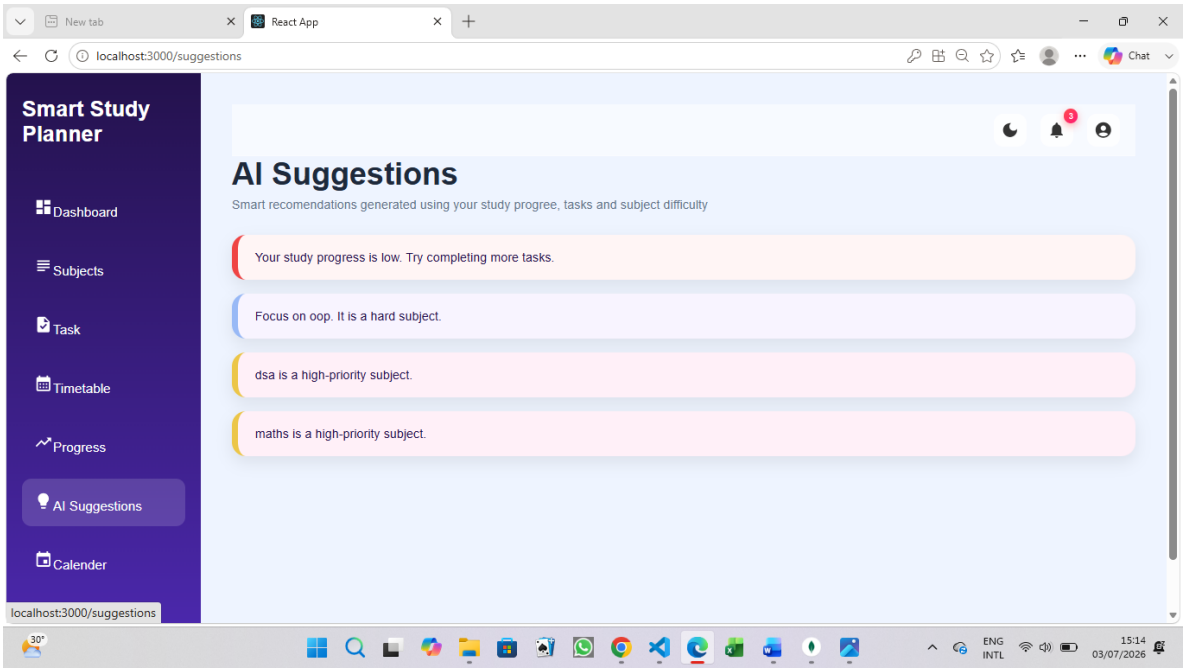


Figure 5.9 AI suggestions page

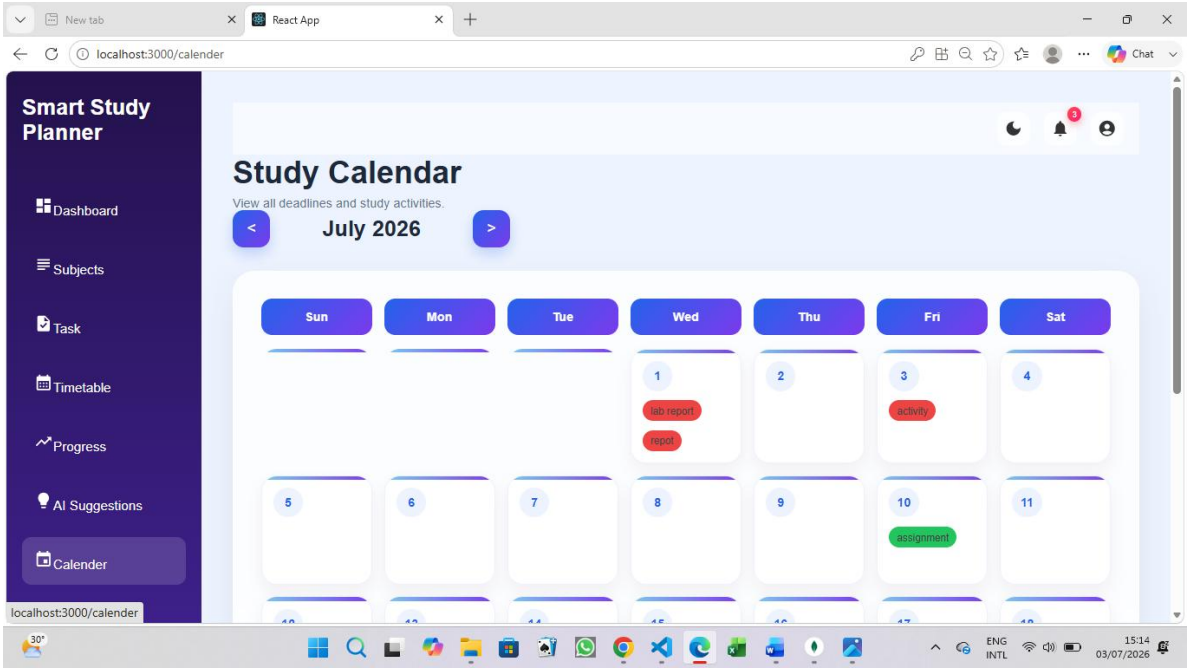


Figure 5.10 Calendar page

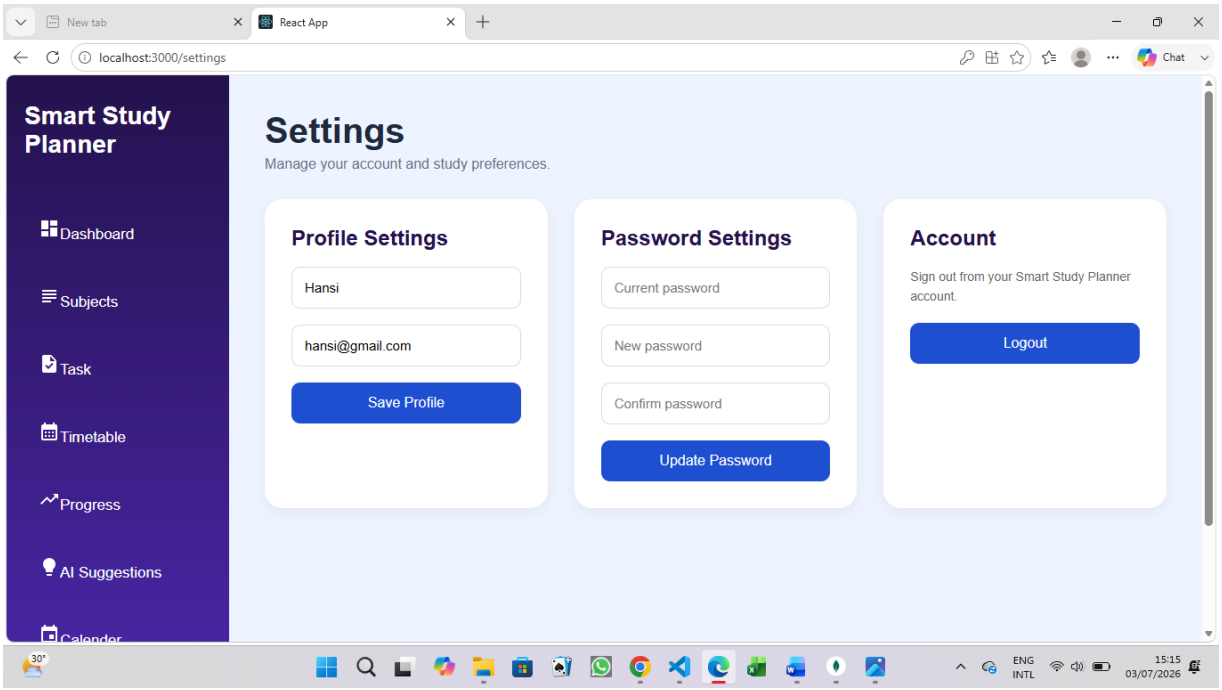


Figure 5.11 Settings page

CHAPTER 6

Testing and Evaluation

6.1 Test Plan

- Testing was conducted in order to detect defects in the software and ensure that all the requirements of the program perform as required.
- Modules that were tested include the following:
  - ✓ Authentication of the user
  - ✓ Management of subjects
  - ✓ Management of tasks
  - ✓ Generation of timetable
  - ✓ Suggestions from AI
  - ✓ Progress Tracking
  - ✓ Calendar
  - ✓ Notifications
- Each module was tested individually before conducting integrated system testing.

6.2 Test Cases

Test Case ID	Description	Input data	Expected outcome	Actual output	Status
TC 001	Register with valid details	Kamal kamal@gmail.com 123456	Account created successfully	Account created and stored in MongoDB	Pass
TC 002	Register with empty fields	Empty Name/ Email/ Password	Validation error displayed	Validation error displayed	Pass
TC 003	Register with existing email	Existing email	Registration blocked	Registration blocked	Pass
TC 004	Register with invalid email	abc123	Invalid email error	Error message displayed	Pass
TC 005	Login with valid credentials	kamal@gmail.com 123456	Login successful	Redirected to Dashboard	Pass
TC 006	Login with wrong password	kamal@gmail.com 111111	Login denied	Invalid password message displayed	Pass
TC 007	Login with unregistered email	test@gmail.com	Login denied	User not found message displayed	Pass

Table 6.2 Test Cases for login

Test Case ID	Description	Input data	Expected outcome	Actual output	Status
TC 008	Add new subject	OOP, hard, high	Subject saved	Subject displayed in table	Pass
TC 009	Add subject with empty field	Subject name empty	Validation error	Error displayed	Pass
TC 010	Edit subject	Change hard → medium	Subject updated	Updated successfully	Pass
TC 011	Delete subject	Click subject	Subject removed	Subject deleted	Pass
TC 012	Add hard subject	ML, hard, high	Subject added	Saved successfully	Pass

TC 013	Add medium subject	ICT, medium, medium	Subject added	Saved successfully	Pass
TC 014	Add easy subject	English, Easy, Low	Subject added	Saved successfully	Pass

Table 6.3 Test Cases for subject

Test Case ID	Description	Input data	Expected outcome	Actual output	status
TC 015	Add task	ML Assignment	Task saved	Displayed in table	Pass
TC 016	Add task with empty fields	Empty task name	Validation error	Error displayed	Pass
TC 017	Edit task	Change deadline	Task updated	Updated successfully	Pass
TC 018	Delete task	Click task	Task removed	Deleted successfully	Pass
TC 019	Set pending status	Status=Pending	Status updated	Pending displayed	Pass
TC 020	Set completed status	Status=Completed	Status updated	Completed displayed	Pass
TC 021	Add task with deadline	2026/07/02	Task saved	Deadline displayed	Pass

Table 6.4 Test Cases for task

Test Case ID	Description	Input data	Expected outcome	Actual output	Status
TC 022	Save study slot	2026/7/1 10-12	Slot saved	Displayed in Saved Study Times	Pass
TC 023	Save slot with empty fields	Empty date	Validation error	Error displayed	Pass
TC 024	Invalid time range	12-10	Validation error	Error displayed	Pass
TC 025	Display saved slots	Existing slots	Show slots	Displayed correctly	Pass
TC 026	Generate timetable	Subjects + Tasks	Timetable generated	Displayed successfully	Pass
TC 027	Generate timetable with tasks	ML Assignment	Task allocated	Displayed correctly	Pass
TC 028	Generate by difficulty	Hard subject	More time allocated	Generated correctly	Pass
TC 029	Generate without subjects	No subjects	No timetable	Empty result displayed	Pass
TC 030	Generate without tasks	No tasks	Subject-only schedule	Generated successfully	Pass

Table 6.5 Test Cases for timetable

Test Case ID	description	Input data	Expected outcome	Actual output	Status
TC 031	Today's Targets	Today's task	Display target	Displayed correctly	Pass
TC 032	Upcoming Tasks	Pending tasks	Display upcoming tasks	Displayed correctly	Pass
TC 033	Study Schedule	Timetable data	Display schedule	Displayed correctly	Pass
TC 034	Display schedule	Completed tasks	Show progress %	Displayed correctly	Pass

Table 6.6 Test Cases for dashboard

Test Case ID	Description	Input data	Expected outcome	Actual output	Status
TC 035	Calculate progress	3/5 completed	60% progress	60% displayed	Pass
TC 036	Subject progress bar	Subject tasks	Show percentage	Displayed correctly	Pass
TC 037	Update progress	Complete task	Progress increases	Updated correctly	Pass

Table 6.7 Test Cases for progress

Test Case ID	Description	Input data	Expected outcome	Actual output	Status
TC 038	Low progress suggestion	Progress <30%	Warning displayed	Suggestion generated	Pass
TC 039	Pending tasks suggestion	>3 pending tasks	Reminder displayed	Suggestion generated	Pass
TC 040	Deadline suggestion	Deadline in 3 days	Deadline in 3 days	Suggestion generated	Pass
TC 041	Excellent progress	Progress >80%	Congratulation message	Suggestion generated	Pass

Table 6.8 Test Cases for suggestions

Test Case ID	Description	Input data	Expected outcome	Actual output	Status
TC 042	Calendar deadline display	Task deadline	Show on calendar	Displayed correctly	Pass
TC 043	Update profile	New name	Profile updated	Updated successfully	Pass
TC 044	Change password	Old + New password	Password changed	Updated successfully	Pass
TC 045	Enable dark mode	Click icon	Dark theme enabled	Enabled successfully	Pass
TC 046	Disable dark mode	Click icon	Light theme enabled	Light theme enabled	Pass
TC 047	Logout	Logout button	Session ended	Redirected to home page	Pass

Table 6.9 Test Cases for Calendar and settings

Test Case ID	Description	Input data	Expected outcome	Actual output	Status
TC 048	Display unread notifications	Existing notification records	Notifications displayed	Notifications displayed correctly	Pass
TC 049	Mark notification as read	Click on notification button	Status updated	Notification successfully marked as Read	Pass
TC 050	Upload PDF	Assignment.pdf	PDF uploaded successfully	Displayed successfully	Pass

Table 6.10 Test Cases for notification

6.3 Types of Testing

- **Unit Testing**

  Tests have been run on unit level to make sure that the individual modules of the Smart Study Planner system work properly before they are integrated together. Every React component and API endpoint in the backend was tested individually. Modules like user registration, user login, subject management, task management, timetable creation, notification generation and progress tracker were tested individually.

- **Integration Testing**

  Testing of the whole software took place once all the modules were completed. The aim of the testing was to check if there was no error in communication among the frontend, backend and MongoDB databases. For instance, when a user enters new tasks through the React interface, the data gets sent to the Express.js API, stored in the MongoDB and then shown immediately in the dashboard and progress modules.

- **Black Box Testing**

  Black box testing involved verifying the functionalities of the system without any concern for the source code inside. Different inputs were provided via the user interface and the output was checked against the expected outcome. It was ensured that all functionalities of the system worked as per the requirements.

- **White Box Testing**

  White-box testing was performed by examining the internal logic of the application. The backend APIs, conditional statements, loops, validation rules and AI suggestion rules were verified to ensure correct execution paths. Source code was reviewed to identify logical errors, unreachable code and incorrect conditions before deployment.

6.4 Results and Analysis

The testing process showed that Smart Study Planner operates in accordance with the defined functional requirements. All modules of the system interacted properly with each other and the application operated in a stable mode during the testing process.

Moreover, the module for generating AI suggestions generated recommendations appropriately considering the prioritization of the subjects, their complexity, number of tasks and deadline. At the same time, the algorithm of generating timetables suggested individual timetables using the provided data about studies.

In conclusion, it can be said that the system performed well in terms of performance, usability and reliability.

CHAPTER 7

Conclusion & Future Work

7.1 Summary of the Project

Smart Study Planner has been successfully created as an academic management system available through the Internet and designed to help students plan their studies better. Subject management, task management, automatic creation of a schedule, AI-based study suggestions, progress tracking, calendar management, notifications and PDF documents management have been successfully combined in one application.

Modern web technologies including React.js, Node.js, Express.js, MongoDB and Mongoose have been successfully utilized during the process of creating this application to ensure its responsiveness and scalability. This application fulfills the requirements set for the project by providing students with the opportunity to manage their time better, track their academic progress and perform their study tasks more effectively.

The tests have proven that all functions of the system are working properly.

7.2 Achievements

The following goals have been accomplished during the development of the application:

- ✓ A system to authenticate users securely.
- ✓ Subject Management & Task Management modules implementation.
- ✓ Creation of automatic personalized studying timetables.
- ✓ A study suggestion module based on artificial intelligence.
- ✓ Progress tracking interaction.
- ✓ An academic calendar system development.
- ✓ Notifications management.
- ✓ PDF file uploads for study materials.
- ✓ User Interface development

7.3 Limitations

While the developed system adequately accomplishes its intended goals, the following are some of the limitations in this system.

- ✓ Only individual users are supported; there are no collaboration functionalities for studying purposes.
- ✓ An internet connection is required to use the app, as there is no offline access feature.
- ✓ Schedules are generated according to the predefined rules, without optimizing them via any algorithm.
- ✓ No email or SMS notifications are available.
- ✓ There is no mobile application functionality.
- ✓ Google Calendar or Microsoft Outlook integration is not provided.
- ✓ The system has undergone testing on a small sample of users.

7.4 Suggestions for Future Improvements

There are several improvements that may be brought to future releases of the Smart Study Planner..

- ✓ Incorporate Machine Learning technology into the system for providing more intelligent study suggestions.
- ✓ Provide an app version for Android and iOS phones.
- ✓ Enable cloud-based synchronization of information on different devices.
- ✓ Enable synchronization with Google Calendar.
- ✓ Enable voice assistant integration.
- ✓ Enable collaborative study planning with groups.
- ✓ Generate detailed analysis and performance reports.
- ✓ Improve the timetable generation algorithm to consider students' study preferences and performances.
- ✓ Email and SMS reminder features.
- ✓ Academic help via chatbots.
- ✓ Pomodoro timer and analytics for the study session.

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